Speaker cue cards

Short prompts to speak naturally - do not read word for word

ARITRI - OPENING

01 Student Success Analytics and Early Warning System

Aritri

- Greet the audience and introduce Aritri and Aston.
- State the project title.
- Explain the aim: identify students who may need support early.
- Make it clear that faculty make the final decision.

NEXT Why is this system needed?

02 Research Problem and Motivation

Aritri

- Student difficulties are often noticed late.
- Useful data already exists, but it is stored in separate records.
- Analytics can organise these signals for earlier faculty attention.

NEXT Explain the main challenges.

03 The Challenge of Timely Intervention

Aritri

- Problems may become visible only after a major assessment.
- Reviewing hundreds of student records manually takes time.
- A Day-40 alert gives faculty time to offer help.
- The alert supports a conversation. It does not predict failure as a fact.

NEXT Move to the research objective and data.

04 Objectives and Data Profile

Aritri

- Main objective: identify possible support needs early.
- Use public learning analytics data for the prototype.
- Use only information available early in the course.

NEXT Introduce OULAD and the Day-40 snapshot.

Speaker cue cards

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HANDOVER: ARITRI TO ASTON

05 Data Profile and Day-40 Snapshot

Aritri

- OULAD is the Open University Learning Analytics Dataset.
- One prepared record represents one student in one module.
- Features include assessment status, scores, VLE activity, credits and previous attempts.
- OULAD does not contain classroom attendance, GPA or backlog fields.

NEXT Show how the data moves through the system.

06 System Workflow and Architecture

Aritri

- Combine assessment and VLE activity up to Day 40.
- Create one clean snapshot for each student and module.
- The model estimates a support tier: Low, Medium or High.
- SHAP explains feature influence. Faculty checks the context.

NEXT Handover to Aston for models and evaluation.

07 Model Comparison and Evaluation

Aston

- Compare Logistic Regression, Decision Tree, Random Forest and XGBoost.
- Select the model using validation results, not accuracy alone.
- Give priority to recall for the High-risk group.
- Also review precision, PR-AUC, ROC-AUC, calibration and faculty workload.

NEXT Show how faculty and students use the results.

08 Dual Dashboard Ecosystem

Aston

- Faculty view: priorities, contributing factors and support options.
- Student view: learning records and links to support resources.
- Students do not see harmful risk labels or ability judgements.
- Every alert requires human verification.

NEXT Present the preliminary results carefully.

Speaker cue cards

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09 Preliminary Prototype Performance

Aston

- Results use historical OULAD 2014J test records.
- High-risk recall is 45.5 percent.
- Combined Medium and High recall is 74.5 percent.
- Higher recall creates more faculty reviews, so workload also matters.
- These results do not prove performance at our institution.

NEXT Explain the safeguards around responsible use.

10 Responsible AI and Governance

Aston

- Educational predictions can influence how people view a student.
- Privacy, fairness and transparency must be part of the design.
- Human responsibility remains central throughout the process.

NEXT Describe the specific responsible-use rules.

11 Responsible Use Principles

Aston

- Faculty verifies every alert before taking action.
- The system cannot change grades, enrolment or course access.
- Only authorised staff should access necessary student data.
- Monitor fairness across student groups.
- SHAP explains model influence, not the cause of a difficulty.

NEXT Finish with the remaining research work.

12 Project Roadmap and Next Steps

Aston

- Compare predictions at Day 20, Day 40 and Day 60.
- Improve recall while keeping review workload manageable.
- Extend subgroup fairness and calibration analysis.
- Complete the research paper in IEEE format.

NEXT Move to the conclusion and questions.

13 Questions and Discussion

Aston, then both

- Conclude: the project helps faculty notice possible support needs earlier.
- The model provides an estimate and explanation.
- Faculty understands the context and chooses any intervention.
- Thank the audience and invite questions.
- Aritri can answer problem, data and workflow questions. Aston can answer model, result and ethics questions.

NEXT If a result is not yet available, say that it is planned future work.